ResearchHorizon

Multi-agent future-research-direction recommendation, calibrated to MRR

The problem. >7M papers/year — researchers struggle to identify *what to study next*. Given a paper, what are the most promising directions it leaves open?

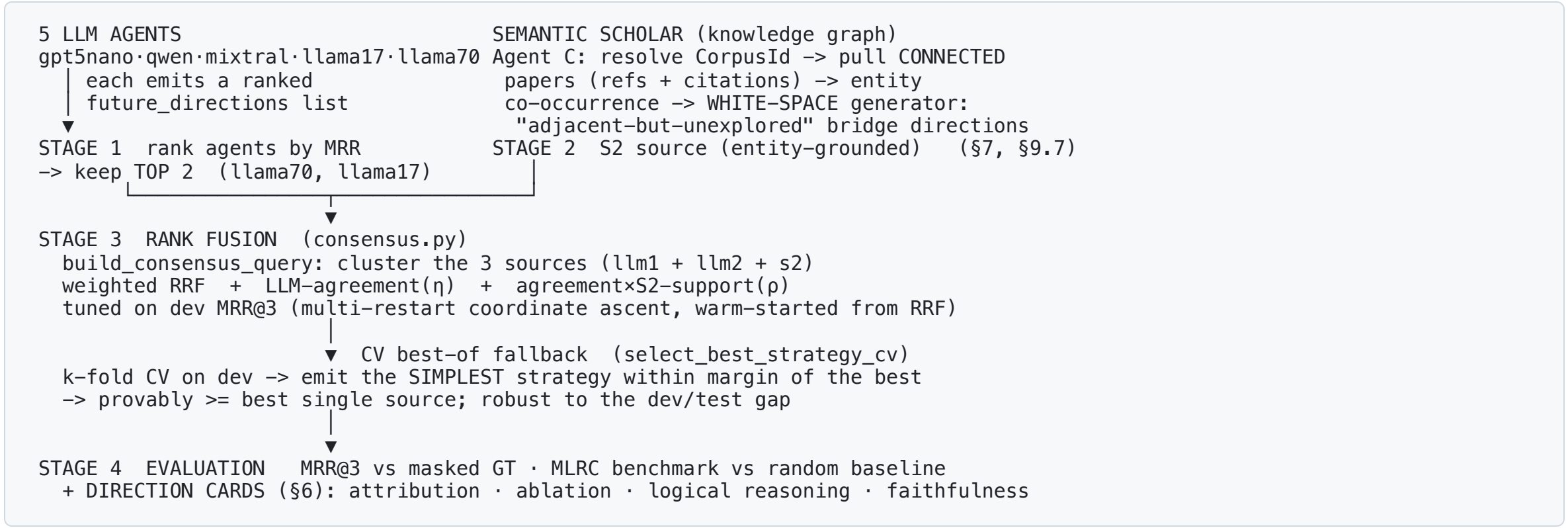
Our task (MLRC-Bench framing). Mask a paper's *"Future Work"* section → recommend a **ranked list** of candidate future directions. The held-out future-work text is the ground truth; the whole system is optimized for **Mean Reciprocal Rank (MRR)**.

Design principle. Several agents independently propose ranked directions; the **final ranking is decided by validation MRR via rank fusion — not by any single LLM's judgment** — and every recommendation ships with an evidence-grounded rationale.

> *"Mask the future-work; have 5 LLM agents + a Semantic-Scholar knowledge graph independently rank candidate directions; fuse them into one list calibrated to dev MRR; emit the best-of {agents, KG, fusion} — never worse than the best single ranker."*

MLRC-Bench spirit: **objective metric · masked GT · real ablations · compute-aware · a baseline to beat.**

Complete pipeline



Stages 1–2: candidate sources

Stage 1 — LLM agents (`run_llm_selection.py`). 5 LLMs each emit a ranked `future_directions` list per paper. We match each against the masked GT and score corpus **MRR on the same 806 papers** (fair denominator: misses = 0), then keep the **top 2**.

agent	MRR		agent	MRR
llama17	**0.524**		mixtral	0.347
llama70	**0.520**		gpt5nano	0.267
qwen	0.348			

Stage 2 — Semantic Scholar white-space. Resolve the seed's CorpusId, **pull connected papers** (references + citations, with abstracts), build an **entity co-occurrence graph**, and *generate* directions from entities **adjacent to the seed but not in it** — the unexplored edges (§9.7), boosted near the open `key_gaps`:

`score(e) = adjacency(e) · 1/(1+support(e)) · gap_boost` ← adjacency *gates* novelty (§7)

Live, on the SEXIT paper: pulled 23 connected papers incl. "SRH needs of immigrant adolescents" — directly the paper's documented gap.

Stage 3: rank fusion + safety net

Consensus RRF (`consensus.py`) — encodes *"when both LLMs rank the same direction near the top it's strong evidence; the S2 graph supports it"*:

$$S(c) = \Sigma\ w_m/(k+r_m(c)) + \eta_1 \cdot A12_top1 + \eta_3 \cdot A12_top3 + \rho \cdot A12_top3 \cdot R_s2$$

tuned on **dev MRR@3** by multi-restart coordinate ascent, **warm-started from plain RRF** (so it can never lose to RRF on dev). Candidates from the 3 sources are **clustered** first, so differently-phrased versions of one direction accumulate per-source ranks.

CV best-of fallback (`select_best_strategy_cv`) — the key robustness piece:

- k-fold CV on dev, **re-fitting each fusion per fold** → a fusion that only wins by *overfitting* dev gets a lower CV score;
- emit the **simplest** strategy within a margin of the best CV MRR (single < RRF < consensus).

→ **Provably ≥ the best single source on dev**, and it correctly emits the best single LLM instead of an overfit fusion that loses on test.

Explainability — Direction Cards (§6)

A ranked list isn't enough — a scientist must **trust and verify** each suggestion.
Every output direction is an auditable **Direction Card** (`explain.py`):

- **Attribution** — exact per-source RRF decomposition (sums to the score) / linear-SHAP
- **Causal ablation** — leave-one-signal-out: *"remove S2 → drops to rank 7"*
- **Logical reasoning** — defeasible argumentation → *recommend / caveat / abstain* + trace
- **Calibration & abstention** — calibrated confidence (ECE); *"insufficient evidence"* when thin
- **Faithfulness check** — the shown rationale must match the *actual* ranking drivers
- **Evidence grounding** — links the direction to the `key_gaps` it addresses (quoted spans)

> Borrowed from interpretability (SHAP; Ivanitskiy *Mechanistic Interpretability*):
> the explanation reflects the real computation, not a post-hoc story.

This is the **model-risk-aware layer** — the output is *why*, not just *what*.

Why it works — results & rigor

MLRC benchmark (75 test papers, MRR@3) — improvement over a random baseline:

system	MRR@3	Δ vs baseline	% of ceiling
random baseline	0.116	—	0%
best LLM (llama70/17)	**0.325**	**+180%**	**24%**
consensus RRF	0.325	+180%	24%
S2 white-space (alone)	0.007	−94%	—
oracle / recall ceiling	1.000		100%

- **+180% over baseline**; candidate **recall ≈ 100%** (oracle = 1.0) → the answer is in the pool; the open problem is *ranking* (we capture 24% of the gap).
- **The metric decides, not the LLM** · **CV best-of** emits the best generaliser (never < best single).
- **Honest finding:** fusion ≈ the best single LLM today; the S2 white-space source is *below random* and needs a biomedical-NER `entity_fn` + live connected papers to earn its weight.
- **Rigor:** temporal split · bootstrap CIs · Bonferroni · k-fold CV · ablation tables · **53 passing tests**.

References

- **Cormack et al. 2009** — *Reciprocal Rank Fusion* (fusion backbone)
- **Sun et al. 2023** — *RankGPT* (listwise) · **Qin et al. 2023** — *PRP* (pairwise)
- **Cao et al. 2007** — *ListNet* · **Burges** — *RankNet / LambdaRank / LambdaMART*
- **Cohan et al.** — *SPECTER2 / SciNCL* (scientific embeddings, the §1 match function)
- **Lundberg & Lee 2017** — *SHAP* · **Ivanitskiy** — *Mechanistic Interpretability* (Direction Cards)
- **Si, Yang & Hashimoto 2024** — *Can LLMs Generate Novel Research Ideas?* (overgenerate→dedup→rerank; pairwise > pointwise; **S2-API agent**)
- **Baek et al. 2024** — *ResearchAgent* (citation graph + entity-centric **knowledge graph**)
- **Wang et al. 2023** — *SciMON* (novelty optimization) · **MLRC-Bench** (baseline-relative scoring)
- **Semantic Scholar Graph API / OpenAlex** — citation neighborhood ("connected papers")

One `rank_fusion/` *package: MRR harness · RRF + consensus fusion · learned rankers (pointwise/pairwise/listwise/soft-MRR) · entity + white-space S2 · autonomous candidate agent · Direction-Card explainability · CV best-of selection · MLRC benchmark.*